JAY A. PATEL

PMRF PhD Scholar | AI & Machine Learning Researcher | LLM & Agent Evaluation

artistjaypatel@gmail.com
me21resch11002@iith.ac.in
Website
Google Scholar
LinkedIn
GitHub
India

PROFILE

PMRF PhD Scholar (CGPA: 10/10) at IIT Hyderabad, researching **wind turbine wakes over complex terrain** using high-fidelity simulations. Also applied **machine learning to CFD problems**, including ML-based prediction of **flow around buildings**. Working as a freelance **AI Model Evaluator** for frontier LLMs, assessing reasoning, factuality, code, and **agent-trajectory** correctness.

EXPERIENCE

Agentic AI Evaluator

Outlier

2026 – Present

- Evaluated **AI agent trajectories** against ground-truth data, detecting **hallucinations, and faked tool calls** across multi-tool workflows.
- Scored agent behavior on **structured rubrics** (correctness, completeness, efficiency) and wrote justifications tracing each claim to the actual trajectory.

AI Model Evaluator

SOUL AI

2025 – Present

- Evaluated **frontier LLM outputs** (**RLHF & SFT**) for reasoning, factuality, instruction adherence, and code/math correctness using structured rubrics.
- Detected **hallucinations** and logical gaps; reviewed **agentic, multi-step reasoning workflows** and gave feedback that improved model response quality.

Generative AI for Turbulence

PhD Research (Ongoing)

2024 – Present

- Developing **deep learning / generative models** to produce realistic turbulent flow fields, combining ML with physics constraints.
- Replacing expensive simulation steps with fast, **learned surrogates** for predictive frameworks.

ML-Based Flow Prediction

PhD Research

2023 – 2024

- Built an **MLP** predicting 3D velocity fields from spatial inputs and SDF encodings; **91% accuracy** vs. high-fidelity CFD data.
- Designed for **low-latency inference** on constrained hardware; validated against physics-based ground truth.

AI & MACHINE LEARNING

LLM Output Evaluation · Reasoning & CoT Review · Hallucination Detection · Factuality Assessment · Generative Models · Neural Networks · Physics-Informed ML · Model Validation · OpenClaw

OTHER SKILLS

High-Performance Computing · Parallel & Distributed Computing · API Integration · Workflow Automation · Browser Automation · Real-Time Systems · Quantum Computing

DATA & ANALYTICS

Large-Scale Data Analysis · Numerical Data Validation · Statistical Post-Processing

PROGRAMMING & TOOLS

Python · C / C++ · Fortran · MATLAB · MPI / OpenMP · Linux / Bash · Git · OpenFOAM · ANSYS Fluent · ParaView

CORE STRENGTHS

Computational Problem Solving · Numerical Methods & PDEs · System Design · Independent Execution · Fast Prototyping

Real-Time Automation System

Independent Project

2023 – Present

- Built an **end-to-end automated decision and execution system** with real-time data ingestion and signal parsing across multiple platforms.
- Integrated platforms via **APIs** and automated API-less systems with **browser automation**; added rule-based logic, retries, error handling, and logging for robustness.

PhD Scholar (PMRF Fellow)

IIT Hyderabad

2021 – Present

- **LES** of turbulent flows over complex terrain; developed an **Immersed Boundary Method (IBM)** solver with wall modelling from scratch.
- Built scalable parallel solvers (**MPI/OpenMP**) on HPC clusters; rigorous verification, validation, and error analysis.

M.Tech Research

IIT Bhubaneswar

2016 – 2018

- Parallel **2D MHD solver in C** (P3DFFT); **fluid–structure interaction** simulations in ANSYS Fluent with UDFs and deforming meshes.

PUBLICATIONS

- Patel, J. A., Maity, A., and Ghaisas, N. S. (2024). *Coupled IBM and wall modelling for high-Re flows over complex terrain*. **Computers & Fluids**, 285, 106457.
- Patel, J. A., Maity, A., and Ghaisas, N. S. (2024). *Topographical influence on wind turbine wakes using LES*. **J. Phys.: Conf. Ser.**
- Ghaisas, N. S., Kethavath, N. N., Patel, J. A., and Mondal, K. (2025). *Wake Models*. Springer Nature Singapore.
- Maity, A., Patel, J. A., and Ghaisas, N. S. (2023). *LES of flow over 2D smooth hills*. **FMFP**.

EDUCATION

PhD, Mech. & Aerospace Engg.

IIT Hyderabad

2021 – Present CGPA: 10/10

M.Tech, Mechanical Engg.

IIT Bhubaneswar

2016 – 2018 CGPA: 9.22/10

GATE, Mechanical Engg.

2016 AIR: 1490

B.E., Mechanical Engg.

GEC Bhavnagar

2011 – 2015 CPI: 7.84/10

WHAT I BRING

- Frontier **AI/LLM evaluation** — know where models fail
- Build **AI models for real systems**, end-to-end
- **Automation & API** engineering experience
- Breadth across **emerging tech** (ML, Quantum, HPC)
- Rigorous, research-grade reasoning

CERTIFICATIONS

Quantum Computing
BIT Mesra (2025)

Turbulence Modeling
IIT Madras (2026)

CFD Workshop
NSM (2023)

Spectral Methods
GIAN (2017)

TEACHING & MENTORING

- **TA** — Adv. CFD, Adv. Computing (IIT Hyderabad, 2021–Present)
- **NPTEL Tutor** — CFD & Comp. Science (2023–2024)
- **Subject Expert** — Fluid Mech. & Math (Chegg, 2021)

LANGUAGES

English

Hindi

Gujarati